Gebze Technical University

Department of Mechanical Engineering

ME 331 System Dynamics and Vibration

Harmonic Force Response

Single-degree-of-freedom systems

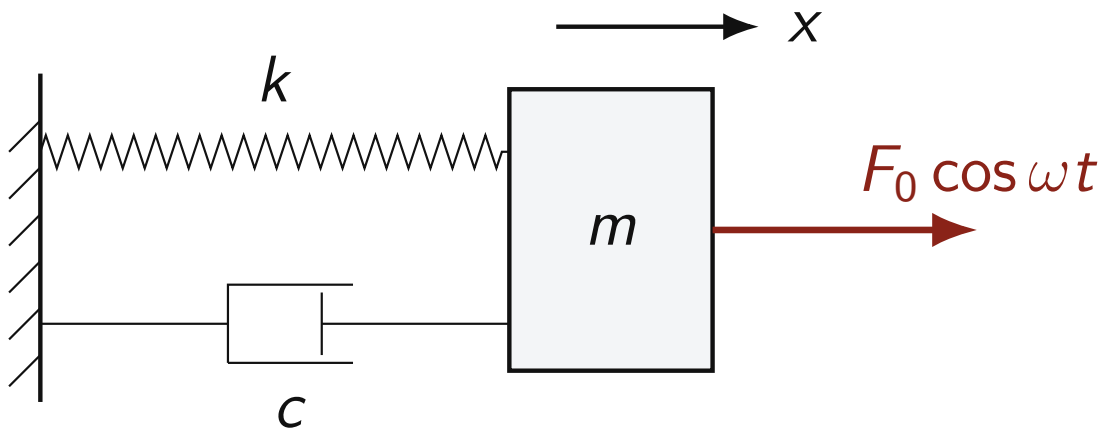
Saeed Lotfan

Today's topics

1. Harmonic excitation
2. Undamped forced response
3. Viscously damped response
4. Frequency-response curves
5. Resonance and damping
6. Worked application and checks

5.1 Harmonic excitation

An applied force drives the mass



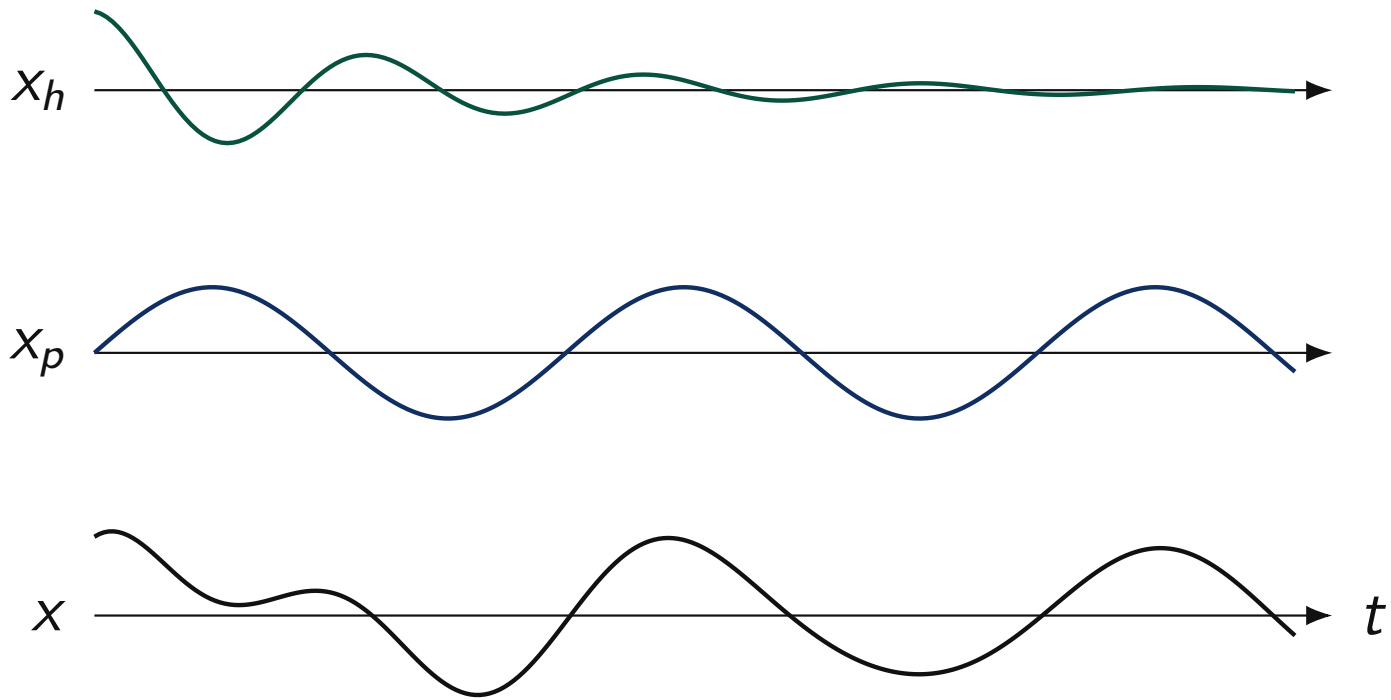
$$m\ddot{x} + c\dot{x} + kx = F_0 \cos \omega t$$

Which frequency is imposed, and which belongs to the system?

5.1 Harmonic excitation

Total response = free part + forced part

$$x(t) = x_h(t) + x_p(t)$$



5.2 Undamped forced response

Apply the initial conditions to the total motion

$$\ddot{x} + \omega_n^2 x = \frac{F_0}{m} \cos \omega t$$

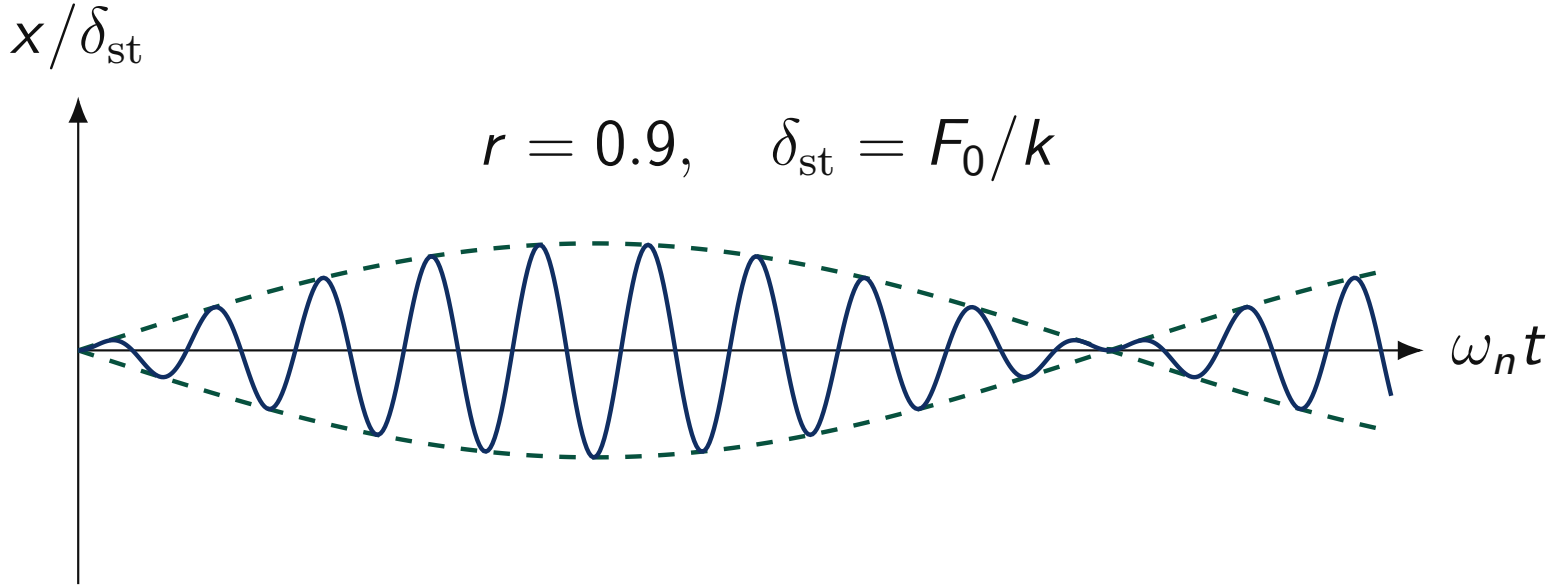
$$x_p = \frac{F_0}{k - m\omega^2} \cos \omega t, \qquad \omega \neq \omega_n$$

Start from rest: $x(0) = 0$, $\dot{x}(0) = 0$.

What changes when $\omega > \omega_n$?

5.2 Undamped forced response

Nearby frequencies produce beating

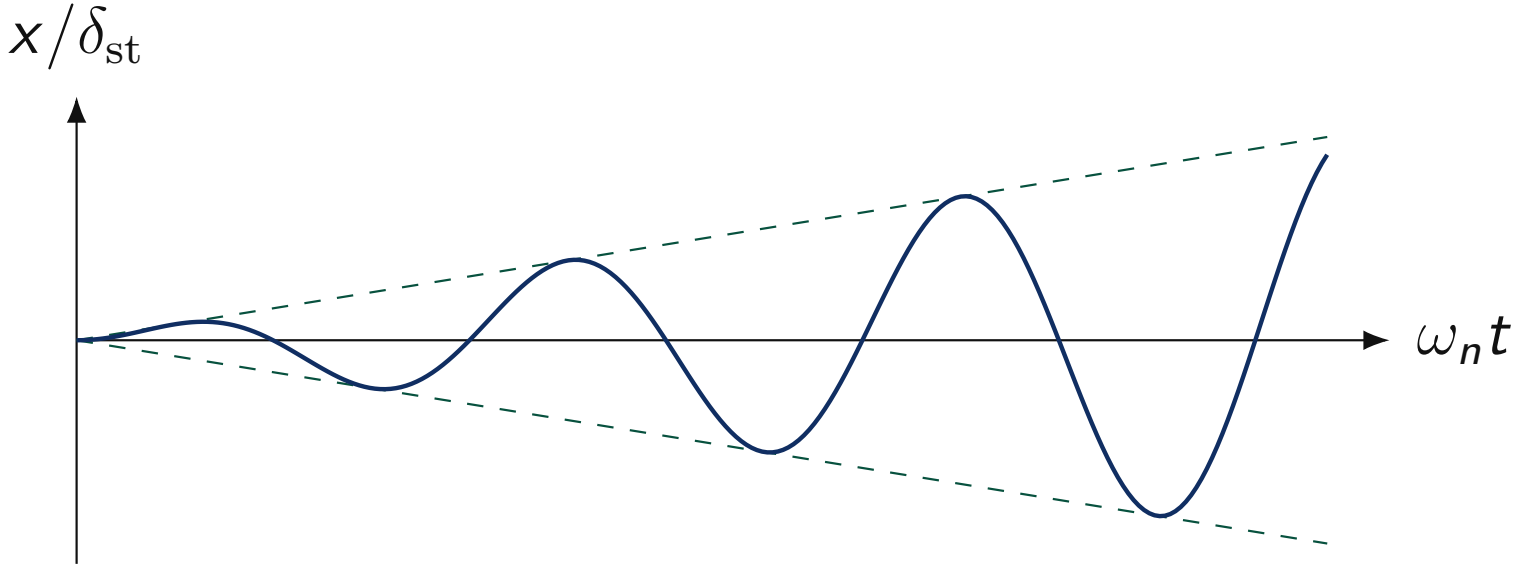


$$\cos\omega t - \cos\omega_n t = -2\sin\frac{(\omega+\omega_n)t}{2}\sin\frac{(\omega-\omega_n)t}{2}$$

5.2 Undamped forced response

Exact resonance: growth with time

$$\omega = \omega_n, \quad x_0 = v_0 = 0: \qquad x(t) = \frac{F_0}{2m\omega_n}\, t \sin \omega_n t$$



5.3 Viscously damped response

Amplitude and phase of the forced motion

$$x_p(t) = X\cos(\omega t - \phi)$$

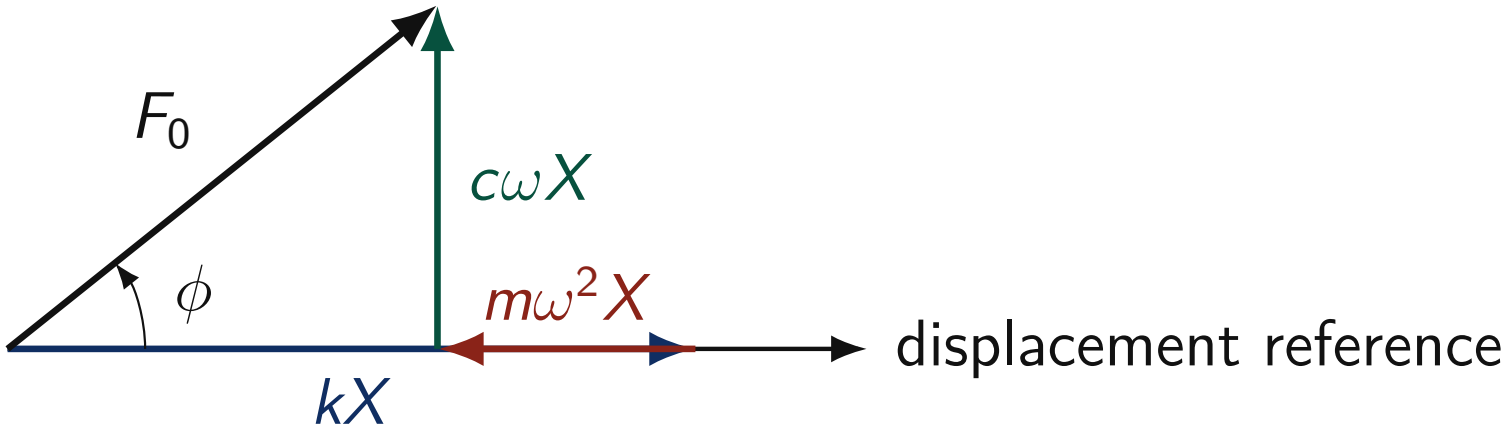
$$X = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$$

$$\phi = \text{atan2}(c\omega,\ k - m\omega^2)$$

Why must the phase quadrant be checked?

5.3 Viscously damped response

A force-vector balance explains phase

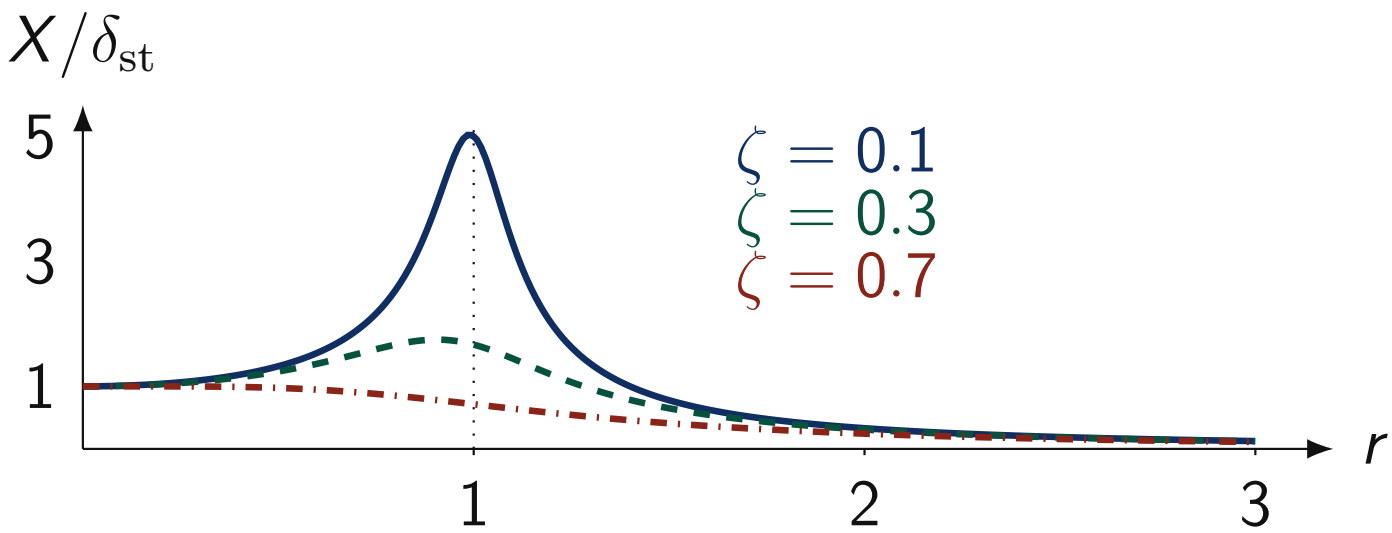


The applied force leads displacement by ϕ.

5.4 Frequency-response curves

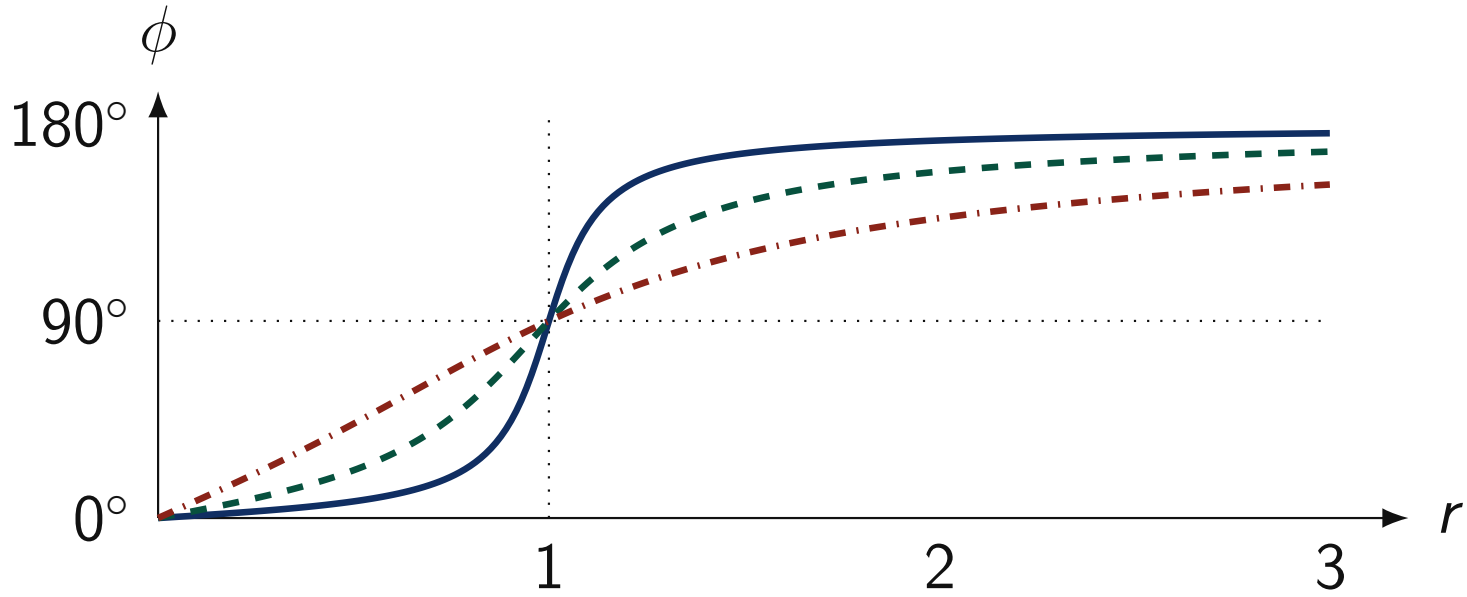
Dynamic magnification

$$r = \frac{\omega}{\omega_n}, \qquad \delta_{\text{st}} = \frac{F_0}{k}, \qquad \frac{X}{\delta_{\text{st}}} = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}$$



5.4 Frequency-response curves

Phase lag changes through resonance



$\zeta = 0.1$ (solid) 0.3 (dashed) 0.7 (dash-dot)

$r \ll 1$: in phase $r = 1$: 90° $r \gg 1$: opposite phase

5.5 Resonance and damping

Peak displacement is not exactly at ω_n

$$\omega_{\text{peak}} = \omega_n \sqrt{1 - 2\zeta^2}, \qquad 0 < \zeta < \frac{1}{\sqrt{2}}$$

$$\left(\frac{X}{\delta_{\text{st}}} \right)_{\text{max}} = \frac{1}{2\zeta\sqrt{1 - \zeta^2}}$$

At $\omega = \omega_n$: $X/\delta_{\text{st}} = 1/(2\zeta)$.

Distinguish ω_d, ω_n, and ω_{peak}.

5.5 Resonance and damping

Complex notation gives the same result

$$F(t) = \Re\{F_0 e^{i\omega t}\}, \qquad x_p(t) = \Re\{\hat{X} e^{i\omega t}\}$$

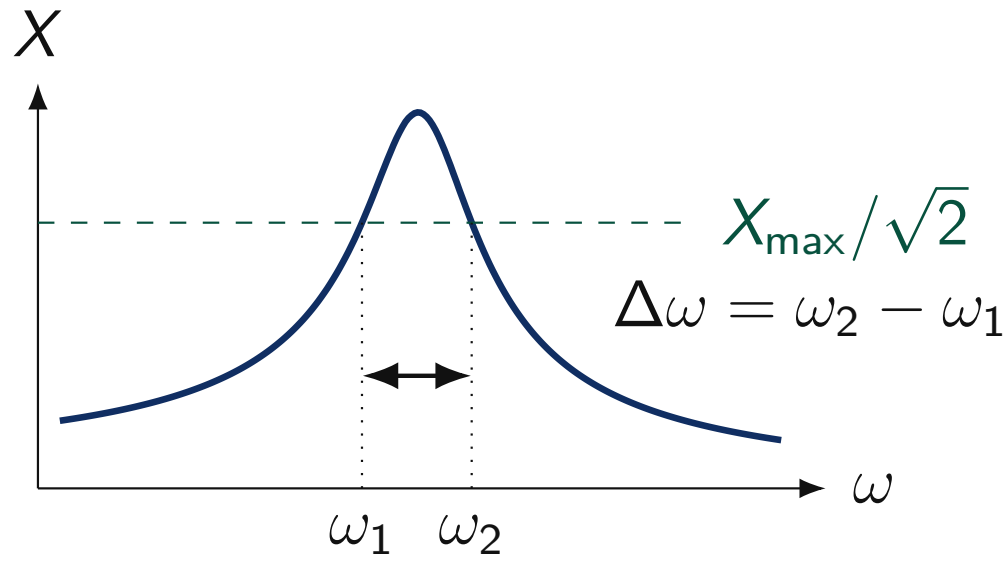
$$(k - m\omega^2 + ic\omega)\hat{X} = F_0$$

$$\hat{X} = \frac{F_0}{k - m\omega^2 + ic\omega} = X e^{-i\phi}$$

Take the real part to recover the physical displacement.

5.5 Resonance and damping

Quality factor and bandwidth: estimate ζ



$$Q \approx \frac{1}{2\zeta} \approx \frac{\omega_n}{\Delta\omega}, \qquad \boxed{\zeta \approx \frac{\omega_2 - \omega_1}{2\omega_n}}$$

Light damping; constant force amplitude. Use $\omega_n \approx \omega_{\text{peak}}$.

5.6 Worked application and checks

Book example: total response

$$m = 10\ \mathrm{kg}, \quad c = 20\ \mathrm{N\,s/m}, \quad k = 4000\ \mathrm{N/m}$$

$$F(t) = 100\cos(10t)\ \mathrm{N}, \quad x_0 = 0.01\ \mathrm{m}, \quad v_0 = 0$$

Find the steady response, then enforce both initial conditions.

Rao, Example 3.3(a)

5.6 Worked application and checks

Before accepting the result

- Do $x(0)$ and $\dot{x}(0)$ match the prescribed values?
- Is the phase in the correct quadrant?
- Does the low-frequency limit give F_0/k?
- Is the resonant case treated separately when $c = 0$?

Whiteboard notes

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